

→ row sampling

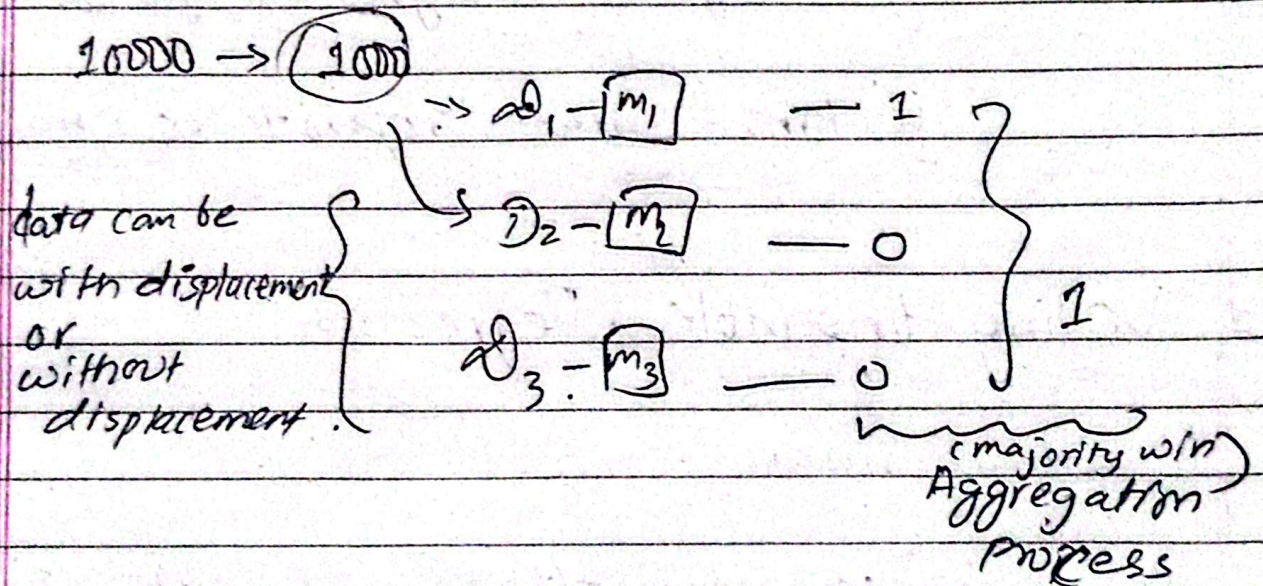
Bagging Ensemble

↓
Bootstrapping

↓
Aggregation

Dataset (10000 rows)

- Take ^{multiple} models homogeneously
- Training each model in different datasets



① Why higher accuracy comes?

⇒ If outer data comes, it splits the dataset, and hence error also splitted. So, variance will be low.

For low bias, we use low bias algorithm at first like DT (non-depth & none), SVM, etc.

③ When to use Bagging.

→ If you have Low Bias High Variance algorithm.

- $df_bag = df_train.sample(8, replace = True)$
 replacement ↓
 one row can repeat.

Boot strap • with replacement.

Pasting.

- row sampling without replacement.

Random Subspaces.

- column sampling.

$df1.sample(2, replace = True, axis = 1)$

Bagging Classifiers.

```
from sklearn.ensemble import BaggingClassifier
```

```
bag = BaggingClassifier(base_estimator = DecisionTreeClassifier,  
    n_estimators = 500,  
    max_sample = 0.25,  
    bootstrap = True,  
    random_state = 42)
```

```
bag.fit(X_train, Y_train)
```

OOB Score [out of bag samples]

when we do sampling, some of the rows (23%) remains untrained,

we can use it by `oob_score = True`
default = False.

And check: `bag.oob_score_` } For just check before answer come.

• It can also be used as testing data too.

Note:

- Bagging gives better prediction than Boosting.
- max-~~sample~~ sample = 25% to 50%.
better results, normally.
- For high dimensional data, use random patches and sub space.
- For hyper parameter tuning, we can do Grid Search CV / Random search CV.

Bagging Regressor.